

COLUMBIA, MO, USA

WMO: 724450

Lat: 38.817N

Lon: 92.218W

Elev: 272

StdP: 98.10

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03945

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.0	-12.8	-20.7	0.6	-15.0	-18.2	0.8	-11.8	12.0	3.8	11.1	2.6	3.8	310	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.9	24.3	33.3	24.4	31.8	23.9	26.4	32.0	25.6	31.1	24.8	30.0	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.8	20.5	29.8	23.9	19.4	29.0	23.1	18.5	27.9	83.6	32.0	79.9	31.1	76.4	30.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.2	8.3	-19.4	37.3	2.6	2.1	-21.3	38.8	-22.8	40.1	-24.3	41.3	-26.2	42.8
			-19.8	27.4	2.6	0.9	-21.7	28.1	-23.2	28.7	-24.7	29.2	-26.6	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-1.1	1.5	7.3	13.2	18.3	23.2	25.4	24.7	20.4	13.9	7.2	1.3		
	DBStd	10.51	6.70	6.62	6.30	5.25	4.45	3.33	3.02	3.07	4.15	5.12	5.55	5.98		
	HDD10.0	1163	347	246	128	27	2	0	0	0	1	21	116	275		
	HDD18.3	2691	602	471	346	167	57	5	0	1	26	155	334	528		
	CDD10.0	2257	3	9	43	123	259	396	476	457	312	141	33	6		
	CDD18.3	745	0	0	2	13	57	151	218	199	88	17	1	0		
	CDH23.3	6947	0	1	22	128	451	1361	2216	1876	752	135	6	0		
	CDH26.7	2758	0	0	1	22	114	511	987	831	261	31	0	0		
(14) Wind	WSAvg	4.1	4.6	4.7	4.9	5.0	4.1	3.7	3.2	3.2	3.4	4.0	4.5	4.4		
(15) Precipitation	PrecAvg	1010	45	50	77	112	122	105	98	96	95	85	71	58		
	PrecMax	1590	151	157	256	297	313	261	308	379	306	279	265	179		
	PrecMin	601	1	3	19	11	33	9	6	5	8	9	9	6		
	PrecStd	219	30	33	44	59	59	59	68	73	70	56	54	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.5	21.7	25.8	29.1	31.4	34.8	38.3	37.6	33.8	29.8	24.1	19.1		
		MCWB	13.1	14.3	16.9	19.0	22.5	23.9	23.8	23.9	23.3	21.7	17.2	15.2		
	2%	DB	14.7	17.9	23.1	26.8	29.4	32.6	35.0	34.9	31.7	26.9	20.9	15.8		
		MCWB	10.5	12.4	15.0	17.4	21.7	24.0	25.0	23.9	22.9	18.9	15.0	12.2		
	5%	DB	11.7	15.0	20.1	24.5	27.8	31.2	33.3	33.2	29.8	24.4	18.3	12.8		
		MCWB	8.0	10.4	13.7	16.5	20.9	23.7	25.0	24.1	22.0	17.9	13.6	9.3		
	10%	DB	8.4	11.6	17.2	22.3	26.0	29.8	31.7	31.2	27.7	22.2	15.9	10.1		
		MCWB	5.4	7.8	12.1	15.4	19.8	23.0	24.5	23.6	20.9	17.0	11.8	7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.7	16.0	18.3	20.5	24.1	26.2	27.5	27.4	25.2	22.6	18.4	16.0		
		MCDB	17.9	19.2	24.2	26.6	29.6	31.7	32.5	32.7	31.3	28.0	22.4	18.7		
	2%	WB	11.5	13.5	16.4	19.0	22.9	25.2	26.6	26.2	24.0	20.9	16.3	13.0		
		MCDB	13.5	16.6	21.1	24.5	28.1	30.7	32.1	31.7	29.5	24.6	19.2	15.4		
	5%	WB	8.2	10.7	14.7	17.7	21.7	24.4	25.8	25.2	23.0	19.4	14.5	10.1		
		MCDB	11.3	14.6	19.0	22.8	26.4	29.6	31.3	30.6	27.6	22.7	17.4	12.1		
	10%	WB	5.3	7.7	12.6	16.4	20.6	23.7	25.1	24.2	22.0	17.8	12.4	7.3		
		MCDB	8.2	11.8	16.6	21.2	24.6	28.7	30.4	29.3	26.3	21.5	15.5	9.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	11.0	11.5	10.7	10.4	10.6	11.1	11.6	11.2	10.2	8.9		
		MCDBR	13.9	14.9	14.5	14.3	12.2	11.7	12.4	13.4	12.8	13.4	13.4	12.7		
	5% WB	MCWBR	10.0	10.3	8.3	7.9	6.3	5.3	4.6	4.9	5.5	7.0	9.0	9.4		
		MCWBR	12.5	13.8	12.6	12.3	11.1	10.7	10.9	11.3	11.0	10.9	11.7	11.6		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.312	0.340	0.386	0.429	0.428	0.447	0.452	0.386	0.345	0.304	0.293		
	taud		2.466	2.418	2.407	2.283	2.225	2.266	2.214	2.214	2.364	2.472	2.529	2.499		
	Ebn at Noon		883	915	919	890	855	853	833	819	858	865	870	862		
	Edh at Noon		80	96	108	130	140	135	141	137	111	90	75	72		
(44) All-Sky Solar Radiation	RadAvg		2.11	2.83	3.78	4.96	5.65	6.41	6.32	5.57	4.82	3.40	2.33	1.81		
	RadStd		0.24	0.29	0.34	0.33	0.49	0.53	0.41	0.31	0.31	0.37	0.24	0.19		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+88		
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82		

Nomenclature: See separate page